

Úloha 2.2

Využijeme nápovědu a zkusme otáčet kanonickou bázi. Pro vektory e_1 a e_2 dostáváme:

$$Re_1 = \begin{pmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos(\beta) \\ \sin(\beta) \end{pmatrix}$$

$$Re_2 = \begin{pmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin(\beta) \\ \cos(\beta) \end{pmatrix}$$

Chceme, aby platilo:

$$\begin{aligned} \langle Re_1, Re_2 \rangle &= (\cos(\beta) \quad \sin(\beta)) \begin{pmatrix} -\sin(\beta) \\ \cos(\beta) \end{pmatrix} \\ &= \cos(\beta) \sin(\beta) - \sin(\beta) \cos(\beta) = 0 \end{aligned}$$

$$\begin{aligned} (Re_1)^T A Re_2 &= (\cos(\beta) \quad \sin(\beta)) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} -\sin(\beta) \\ \cos(\beta) \end{pmatrix} \\ &= (\cos(\beta) \quad \sin(\beta)) \begin{pmatrix} b \cos(\beta) - a \sin(\beta) \\ c \cos(\beta) - b \sin(\beta) \end{pmatrix} \\ &= b \cos^2(\beta) - a \sin(\beta) \cos(\beta) + c \sin(\beta) \cos(\beta) - b \sin^2(\beta) \\ &= b \cos(2\beta) + \frac{1}{2}(c - a) \sin(2\beta) = 0 \end{aligned}$$

$$\operatorname{tg}(2\beta) = \frac{2b}{a - c}$$

$$\beta = \frac{1}{2} \operatorname{arctg} \left(\frac{2b}{a - c} \right)$$